

Paper III — Mass Without a Higgs Field: Chiral- $SU(2)$ β -Gauging, the Spherical Mass Shell, and the Koide Relation

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Notebook source material: M. Ludwig (2007), *Physics Notes*, pp. 59–60 (“Complex mass”), pp. 73–74 (helical mass)

Series: “A Universe in a Bottle” — Paper III of XI. Builds on Paper I (spherical mass shell) and is the mass foundation for Papers VI, IX, X, XI.

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Abstract

We show that fermion mass on the BCC quantum-cellular-automaton vacuum is generated without a Higgs field. The mechanism is Ludwig’s chiral- $SU(2)$ complex-mass coupling: replacing the scalar Dirac mass im by a local complex phase $im e^{i\theta(\mathbf{x})}$ is unitarily equivalent to gauging the Dirac β -matrix, and promoting the phase to a local $SU(2)$ field $U(\mathbf{x})$ endows the mass step with an exact chiral $SU(2)_L$ gauge symmetry — the weak isospin group — verified by a Ward identity at 1.1×10^{-17} . The field $U(\mathbf{x})$ plays the role of the Higgs vacuum direction but is a pure gauge degree of freedom: no physical scalar appears, the dispersion is independent of the mass phase θ , yet a mass gap forms with vanishing condensate. We connect this to the spherical-Pythagorean mass shell of Paper I ($E^2 = p^2 c^2 + m^2 c^4$ as a continuum limit), and to the constituent-mass scale via a self-consistent Nambu–Jona-Lasinio (NJL) gap equation with critical coupling $G_c \Lambda^2 = \pi^2/6$ that reproduces the measured light-meson sector. Finally we derive the long-standing Koide relation $Q = \sum m / (\sum \sqrt{m})^2 = 2/3$ for the charged leptons as an *equipartition* condition on a cubic vector $\sqrt{m}$, itself explained as a Cooper-pair bilinear ($m = y^2$). All structural results are verified to machine precision.

1. Introduction

In the Standard Model, fermion masses arise from Yukawa couplings to a Higgs doublet whose vacuum expectation value (VEV) breaks electroweak symmetry. The Higgs sector is the least economical part of the SM: it adds a fundamental scalar, a potential tuned by hand, and a Yukawa matrix of free parameters. The notebook that seeds this programme proposed a different route in 2007 — “complex mass” — and a central design decision of the project is to adopt it in place of the Higgs (Paper I, §1.1).

This paper develops that route into a complete mass mechanism. The thesis is three-fold. (i) The *origin* of mass is a chiral rotation of the lattice spinor that carries weak

isospin as a gauge symmetry, requiring no scalar (§2-3). (ii) The *mass shell* it produces is the spherical-Pythagorean identity of Paper I, whose continuum limit is Einstein's $E^2 = p^2 c^2 + m^2 c^4$ (§4). (iii) The *magnitudes* — the constituent mass scale and the charged-lepton mass ratios — emerge from the lattice's own pairing dynamics, the latter as the Koide relation (§5-6).

2. The complex-mass coupling

In the Weyl representation the standard Dirac mass couples the left- and right-handed two-spinors η, χ via

$$M_0 = \begin{pmatrix} 0 & im I_2 \\ im I_2 & 0 \end{pmatrix}.$$

Ludwig's proposal replaces the scalar coupling im by a **local complex phase**:

$$M(\theta) = \begin{pmatrix} 0 & im e^{i\theta} I_2 \\ im e^{-i\theta} I_2 & 0 \end{pmatrix}.$$

This is equivalent to gauging the Dirac β -matrix: with $U = \text{diag}(1, e^{i\theta})$,

$$\beta_g = (U \sigma_1 U^\dagger) \otimes I = \cos \theta \sigma_1 + \sin \theta \sigma_2,$$

a rotation in the σ_1 - σ_2 plane. Promoting $\theta \rightarrow \theta(\mathbf{x}, t)$ makes this a local gauge transformation.

2.1 The mass step and its exactness

Writing $c_m = \cos(m \Delta t)$, $s_m = \sin(m \Delta t)$, the single-flavour mass step is

$$\begin{pmatrix} \eta' \\ \chi' \end{pmatrix} = \underbrace{\begin{pmatrix} c_m & is_m e^{i\theta} \\ is_m e^{-i\theta} & c_m \end{pmatrix}}_{M(\theta)} \begin{pmatrix} \eta \\ \chi \end{pmatrix}.$$

The matrix $M(\theta) = c_m I + is_m A$ with $A = \begin{pmatrix} 0 & e^{i\theta} \\ e^{-i\theta} & 0 \end{pmatrix}$ Hermitian and $A^2 = I$, so $M^\dagger M = I$ exactly for every $\theta(\mathbf{x})$. Norm drift over 50 random- θ steps is 1.3×10^{-15} (machine precision). At $\mathbf{k} = 0$ the step is a pure rotation of the chirality pair at rate $\Omega_{\text{rest}} = \arcsin m$ per tick — the rest leg of the spherical mass shell (Paper I, §5).

3. Chiral $SU(2)_L$ as a gauge symmetry of the mass step

For an isospin doublet $(\nu, e)_L$, replace the scalar phase $e^{i\theta}$ by $U(\mathbf{x}) \in SU(2)$ acting on the isospin index while leaving the spin index untouched:

$$\eta' = c_m \eta + i s_m (U \otimes I_{\text{spin}}) \chi, \quad \chi' = i s_m (U^\dagger \otimes I_{\text{spin}}) \eta + c_m \chi.$$

The full mass operator is $M_{SU(2)} = \cos I_8 + i \sin \begin{pmatrix} 0 & U \otimes I \\ U^\dagger \otimes I & 0 \end{pmatrix}$, unitary by the same argument as (2.4).

3.1 The Ward identity (the central result)

Let $V(\mathbf{x}) \in SU(2)$ act **only on the left-handed η sector**. Then the mass step satisfies, exactly,

$$V(\mathbf{x}) \cdot \text{mass_step}(\psi; U) = \text{mass_step}(V(\mathbf{x}) \cdot \psi; V(\mathbf{x}) U).$$

That is, simultaneously transforming $\eta \rightarrow V\eta$, $U \rightarrow VU$, and $\chi \rightarrow \chi$ (unchanged) leaves the mass step invariant. The symmetry group is $SU(2)_L$ — precisely the weak isospin group of the SM — derived from β -gauging alone, with **no Higgs field**. The Ward-identity residual is 1.1×10^{-17} (the key test, Finding F27). Chirality is exact: under the $SU(2)_L$ transform the right-handed χ is unchanged to bit-for-bit 0, while η transforms non-trivially.

3.2 $U(\mathbf{x})$ is the Higgs vacuum direction, but pure gauge

In the SM the Higgs VEV $\langle \Phi \rangle$ selects which doublet component couples to the right-handed singlet. Here $U(\mathbf{x})$ performs the same selection: starting from a pure ν_L state, $U = I$ excites ν_R , $U = i\sigma_1$ excites e_R , a 45° mix excites both equally. But U is a **pure gauge degree of freedom**, not a physical boson:

- **Dispersion independence.** The eigenvalues of the mass operator are independent of θ to 3.3×10^{-16} across $\theta \in \{0, \pi/3, \pi/2\}$ — θ carries no physical information.
- **Mass gap without condensate.** With $U = I$ (no VEV, no scalar) a mass gap still forms: with $m \neq 0$ the right-chirality fraction grows to $N_R = 0.82$ after 80 ticks, while $m = 0$ stays pure-left to 4.0×10^{-15} .
- **Correct quantum numbers.** A pure ν_L state carries $T_3 = +\frac{1}{2}$ to 1.1×10^{-16} .

3.3 Status as a known mass mechanism

The construction is the lattice realisation of the non-Abelian Stueckelberg / Kunimasa-Goto mass term: a gauge-invariant mass generated without a physical scalar. The continuum no-go theorems on renormalisability of massive Yang-Mills do not bind here, because the lattice carries a built-in UV cutoff (the cell size a). The phase that would have been the Higgs is eaten as the Stueckelberg/longitudinal mode (Paper

VI; hypercharge in Paper X). The hypercharge $U(1)_Y$ rides on the *same* field $U(\mathbf{x})$ that carries the mass phase, by absorbing the chiral hypercharge difference $\Delta Y = Y_L - Y_R$ (Finding F41); this is what lets the model dispense with the Higgs scalar entirely rather than merely relocating it.

4. The mass shell: spherical-Pythagorean composition

The mass rotation of §2–3 composes with the kinetic (Weyl) rotation of Paper I by the **spherical law of cosines**. For the exact-QCA Dirac propagator $D_{\mathbf{k}}$ (Paper I, Eq. 5.1) with admissibility $n^2 + m^2 = 1$, $n = \sqrt{1 - m^2}$:

$$\cos \Omega_{\text{Dirac}}(\mathbf{k}, m) = \sqrt{1 - m^2} \cos \omega_{\text{kin}}(\mathbf{k}) = \cos \Omega_{\text{rest}}(m) \cos \omega_{\text{kin}}(\mathbf{k}),$$

with $\Omega_{\text{rest}} = \arcsin m$. Small-angle expansion yields $\Omega_{\text{Dirac}}^2 = m^2 + \omega_{\text{kin}}^2 + O(\text{lattice}^4)$, i.e.

$$E^2 = m^2 c^4 + p^2 c^2 + O(\text{lattice}^4).$$

Einstein’s mass shell is the continuum limit of a spherical-Pythagorean identity whose two legs are the mass-rotation rate and the kinetic-rotation rate. This vindicates a 2007 notebook construction (pp. 73–74) in which mass was modelled as a particle confined to a helical trajectory, $c^2 = v_{\text{eff}}^2 + (2\pi\nu r)^2$ — a Pythagorean split of the lattice light speed into a propagation leg and a rest-rotation leg. The notebook dismissed it for the wrong reason (the high-energy limit $v \rightarrow c$, which is in fact correct for massive particles); the exact lattice identity is the spherical form, with the helical decomposition as its Euclidean limit. Mass, in this picture, is literally *confined rotation*: a massless quantum forced to turn in place rather than stream forward. This is the same statement that lets Paper VII treat all mass as confined field-rotation energy and dispense with a separate gravitational source.

5. The constituent-mass scale: a self-consistent NJL gap

The chiral mass step says *how* mass appears; the *magnitude* of the dynamical (constituent) mass is fixed by the lattice’s pairing dynamics, modelled as a Nambu–Jona-Lasinio (NJL) four-fermion contact interaction. A single coupling G both generates the constituent mass via the gap equation and, summed in the $q\bar{q}$ ladder (RPA), fixes the meson poles (Finding F77).

Gap equation (Hartree, $N_c = 3$, $N_f = 2$):

$$M = m_0 + 4G N_c N_f M I_1(M), \quad I_1(M) = \frac{1}{2\pi^2} \int_0^\Lambda \frac{p^2 dp}{\sqrt{p^2 + M^2}},$$

with a finite **critical coupling**

$$G_c \Lambda^2 = \frac{\pi^2}{N_c N_f} = \frac{\pi^2}{6} = 1.644934$$

below which chiral symmetry is unbroken. In the chiral limit the pion is the exact Goldstone boson, and the scalar partner sits at threshold, $m_\sigma = 2m_c$ (residual 2.3×10^{-14}). The canonical SU(2) fit ($\Lambda = 651.5$ MeV, $G\Lambda^2 = 2.10$, $m_0 = 5.5$ MeV) reproduces the measured QCD numbers, certifying the normalisation:

quantity	model	measured	resid
m_c	311 MeV	$\sim \$325$	4.2%
f_π	92.6 MeV	92.4	0.2%
m_π	140.5 MeV	135-138	4.1%
$\langle \bar{q}q \rangle^{1/3}$	-249 MeV	~ -250	0.4%

A by-product (relevant to any composite-Higgs speculation) is that the self-consistent scalar always obeys $m_\sigma/2m_c \geq 1$, so the dynamical-mass mechanism cannot produce the sub-threshold binding ($m_H/2m_t = 0.363$) a $t\bar{t}$ composite Higgs would require — a structural near-no-go. The same NJL machinery, promoted to a dynamical pion, supplies the long-range nuclear force in Paper XI.

6. The charged-lepton mass ratios: the Koide relation

6.1 Koide as cubic-vector equipartition

The generation structure (Paper IX) places the three charged leptons in the cubic vector irrep T_{1u} of O_h . Define the vector $\mathbf{s} = (\sqrt{m_e}, \sqrt{m_\mu}, \sqrt{m_\tau})$ and split it into its democratic (A_{1g} , along $\hat{n} = (1, 1, 1)/\sqrt{3}$) and traceless (T_{1u}) parts. The angle θ between $\mathbf{s}$ and $\hat{n}$ satisfies the **exact identity**

$$\cos^2 \theta = \frac{(\sum_a \sqrt{m_a})^2}{3 \sum_a m_a} = \frac{1}{3Q}, \quad Q \equiv \frac{\sum_a m_a}{(\sum_a \sqrt{m_a})^2}.$$

Koide's empirical $Q = \frac{2}{3}$ is therefore exactly the **equipartition condition** $\cos^2 \theta = \frac{1}{2}$, i.e. $\theta = 45^\circ$, i.e. the cubic-scalar and symmetry-breaking parts of $\sqrt{m}$ carry equal weight. The measured charged leptons give $Q = 0.6666605$ (0.91σ from $\frac{2}{3}$), $\theta = 44.99974^\circ$, ratio $|A_{1g}|^2/|T_{1u}|^2 = 1.00002$.

6.2 Predictivity

Because $Q = \frac{2}{3}$ plus two masses overdetermines the third, the relation is predictive:

$$m_\tau^{\text{pred}} = 1776.97 \text{ MeV} \quad \text{vs} \quad m_\tau^{\text{PDG}} = 1776.86 \pm 0.12 \text{ MeV} \quad (6.1 \times 10^{-5}).$$

A threefold (Z_3 , cube-body-diagonal) parametrisation $\sqrt{m_a} = M_0[1 + \sqrt{2}\cos(\delta + 2\pi a/3)]$ reconstructs m_τ to 4.4×10^{-5} , with the $\sqrt{2}$ amplitude being exactly the equipartition condition. The angle map $Q(\phi) = 1/(3\cos^2\phi)$ (Finding F80) makes the correspondence $45^\circ \leftrightarrow Q = \frac{2}{3}$ exact.

6.3 Why $\sqrt{m}$: the Cooper-pair bilinear

The use of $\sqrt{m}$ rather than m is explained by the same Cooper-pair premise that makes the photon a bound pair (Paper II) and the would-be Higgs a condensate. If a fermion's mass is set by a pair condensate, it is a *bilinear* in the constituent amplitude y :

$$m_a \propto \langle y_a y_a \rangle = y_a^2 \implies \sqrt{m_a} = y_a = \text{the } T_{1u} \text{ vector component on axis } a.$$

Koide is then a statement about the amplitude y , not the mass — precisely because mass is quadratic in the fundamental pairing amplitude (Finding F78). The data independently single out the bilinear: the participation ratio $(\sum m^s)^2 / \sum m^{2s}$ is the clean rational $3/2$ only at $s = \frac{1}{2}$ (measured 1.500014; neighbours 1.834 at $s = \frac{1}{3}$, 1.119 at $s = 1$).

6.4 Honest limits

The cubic geometry fixes the *form* (three orthorhombic generations; $\sqrt{m}$ as cubic vector; the equipartition identity (6.1)) and the charged leptons sit on it to 10^{-5} . The amplitude $\sqrt{2}$ itself ($Q = \frac{2}{3}$ exactly) is **not yet derived** from the QCA rule: the cube's symmetric dynamics give the degenerate floor $Q = \frac{1}{3}$, so equipartition is a critical/maximal-breaking condition that an explicit ingredient must impose (Finding F78). A recurring $\sqrt{2}/45^\circ$ in both the constituent stability bound (F73) and the generation vector hints at a single saturation mechanism; Finding F80 identifies the *selector* of the critical sector as electromagnetism (only the colour-free, charged sector lands on 45°), while flagging that perturbative EM is $\sim 340\times$ too weak to *drive* the rotation. The up- and down-type quarks ($Q = 0.85, 0.73$) do not equipartition; why only the charged leptons do is treated further in Paper X.

7. Verification summary

Result	Section	Residual / value
Mass-step unitarity (random θ)	§2.1	1.3×10^{-15}
$SU(2)_L$ Ward identity (3.2)	§3.1	1.1×10^{-17}
Chirality exact (χ unchanged)	§3.1	0 (bit-for-bit)
Dispersion independence of θ	§3.2	3.3×10^{-16}
Mass gap without Higgs (N_R)	§3.2	0.82 (vs 4.0×10^{-15} at $m = 0$)
Spherical-Pythagorean shell (4.1)	§4	3.3×10^{-16}
NJL Goldstone $m_\sigma = 2m_c$ (chiral)	§5	2.3×10^{-14}
Koide Q (charged leptons)	§6.1	0.6666605 (0.91σ)
Koide m_τ prediction	§6.2	6.1×10^{-5}

Underlying findings: F27 (chiral $SU(2)$ complex mass), F46 (spherical-Pythagorean shell), F77 (NJL gap+RPA), F78 (Koide amplitude from Cooper pair), F80 (45° / EM selection), F76 (generation hierarchy), F44 (rank-1 Stueckelberg $m_A = 0$), F41 (hypercharge on $U(\mathbf{x})$).

8. Discussion and relation to the series

This paper supplies the mass mechanism that the rest of the matter sector inherits. Paper VI (weak force) gauges the $SU(2)_L$ symmetry derived in §3 with dynamical W bosons. Paper IX (fermion sector) uses the spherical mass shell (§4) and the cubic generation structure (§6) to fix the generation count and hierarchy. Paper X (leptons) adds the right-handed-neutrino Majorana mass and see-saw on the same Higgs-free footing, and develops the Koide/EM-selection story. Paper XI (baryons) uses the NJL constituent-mass scale (§5) and its dynamical pion.

The single conceptual claim unifying all of this — that mass is a *rotation* (the rest leg of the spherical triangle) sourced by a pure-gauge chiral coupling rather than by a fundamental scalar — is what makes the model Higgs-free without sacrificing any verified mass phenomenology.

References

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4. M. Ludwig, *Physics Notes* (2007), pp. 59–60 (“Complex mass”), pp. 73–74 (helical mass construction).
5. Project findings: F27, F41, F44, F46, F76, F77, F78, F80.

Companion papers: I (substrate/mass shell), VI (weak force), IX (fermions), X (leptons), XI (baryons).